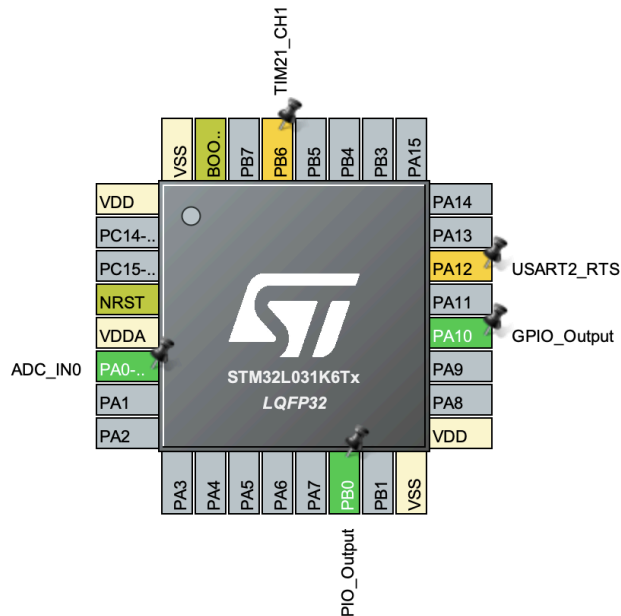


Pin Configuration



GPIO Pin Configuration

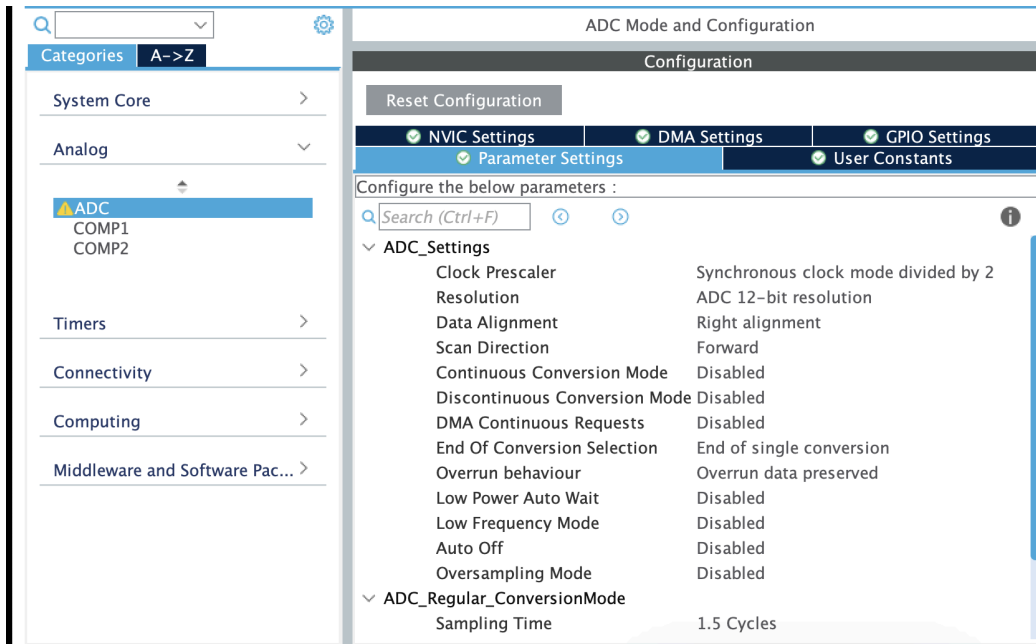
GPIO pins were configured as digital outputs for controlling the LED and buzzer alarm indicators. These outputs are activated whenever environmental conditions exceed predefined thresholds.

UART Configuration

UART communication was configured to transmit sensor readings and system status messages to the serial monitor for real-time monitoring and debugging.

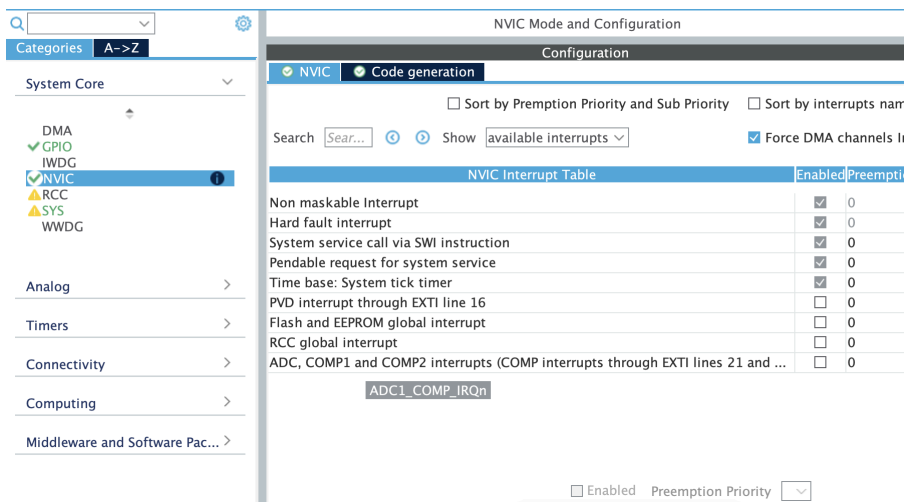
Timer Configuration

Timer peripherals provide periodic timing events for sensor sampling and alarm management. In a fully optimized implementation, timer interrupts can replace software delays.



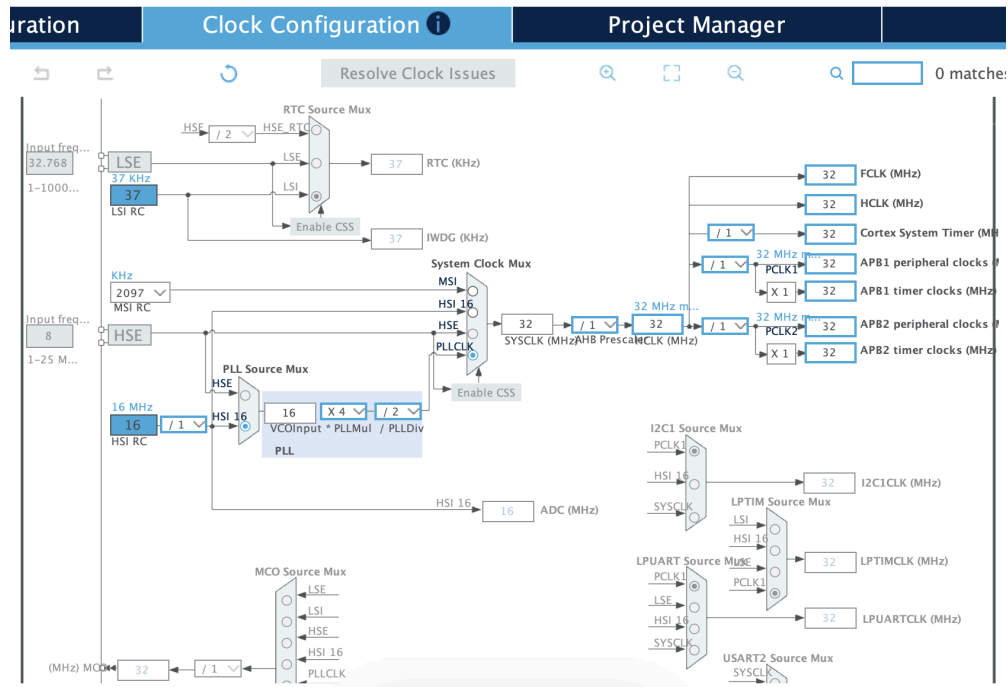
ADC Configuration for LDR Sensor

The ADC peripheral converts the analog voltage produced by the LDR sensor into a digital value that can be processed by the STM32 microcontroller.



Interrupt (NVIC) Configuration

Interrupts allow the processor to respond immediately to timer events without continuously polling the peripherals.



STM32 Clock Configuration

The system clock was configured to operate at 32 MHz using the internal high-speed oscillator (HSI), providing sufficient performance for sensor acquisition and alarm processing.